

Exoplanets – Nested Sampling

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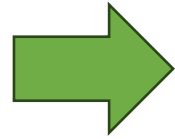
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Evidence and Model Selection

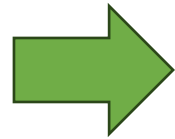
Bayes' Theorem with explicit dependance on model $\mathcal{M}_0$



$$P(\boldsymbol{\theta}|\mathbf{D}, \mathcal{M}_0) = \frac{P(\mathbf{D}|\boldsymbol{\theta}, \mathcal{M}_0) P(\boldsymbol{\theta}, \mathcal{M}_0)}{P(\mathbf{D}, \mathcal{M}_0)}$$

$$= \frac{\mathcal{L}(\boldsymbol{\theta}) \pi(\boldsymbol{\theta})}{P(\mathbf{D}, \mathcal{M}_0)}$$

Bayes' Factor: quantifies the support of one model over the other



$$K = \frac{P(\mathbf{D}|\mathcal{M}_1)}{P(\mathbf{D}|\mathcal{M}_0)} = B_{10}$$



[Kass & Raftery 1995](#)

$$P(\mathbf{D}|\mathcal{M}_0) = \int_{\Omega_{\boldsymbol{\theta}}} \mathcal{L}(\boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta} = \mathcal{Z}_0$$

$$B_{10} = \frac{\mathcal{Z}_1}{\mathcal{Z}_0}$$

$$\ln(B_{10}) = \ln(\mathcal{Z}_1) - \ln(\mathcal{Z}_0)$$

$2 \log_e(B_{10})$	(B_{10})	Evidence against H_0
0 to 2	1 to 3	Not worth more than a bare mention
2 to 6	3 to 20	Positive
6 to 10	20 to 150	Strong
>10	>150	Very strong

MCMC & Nested Sampling

Bayes' theorem

Posterior

$$P(\theta|D, \mathcal{M}_0) = \frac{P(D|\theta, \mathcal{M}_0) P(\theta, \mathcal{M}_0)}{P(D, \mathcal{M}_0)}$$

Likelihood * Prior

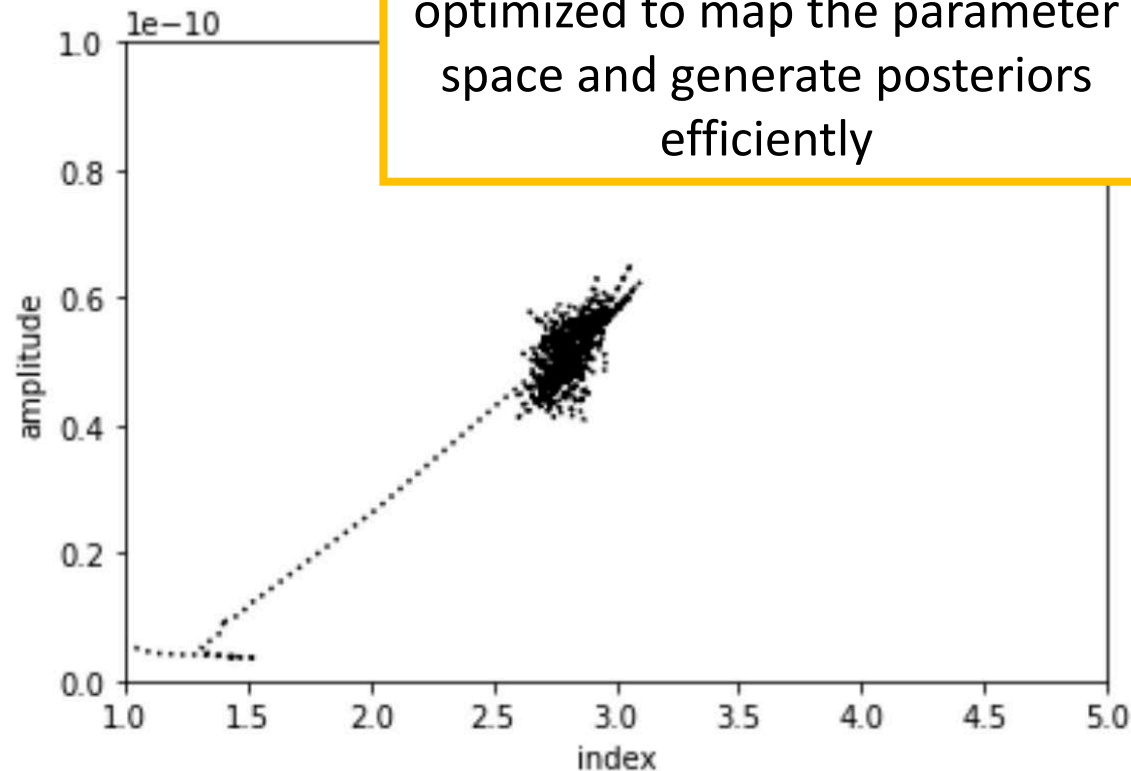
Evidence

MCMC: [Goodman & Weare](#)

NS: [Skilling 2006](#)

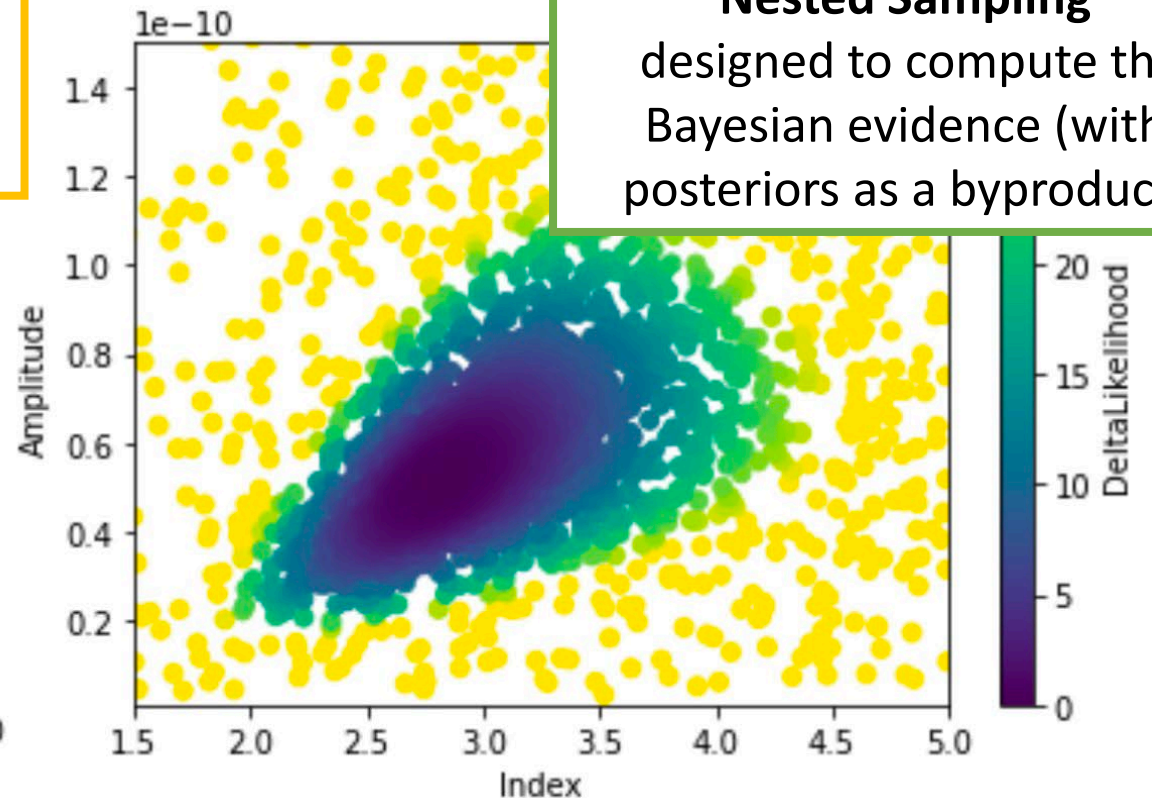
Markov Chain Monte Carlo

optimized to map the parameter space and generate posteriors efficiently



Nested Sampling

designed to compute the Bayesian evidence (with posteriors as a byproduct)



Priors in NS

Parameters X are uniformly sampled in the range $[0,1]$:
unit hypercube

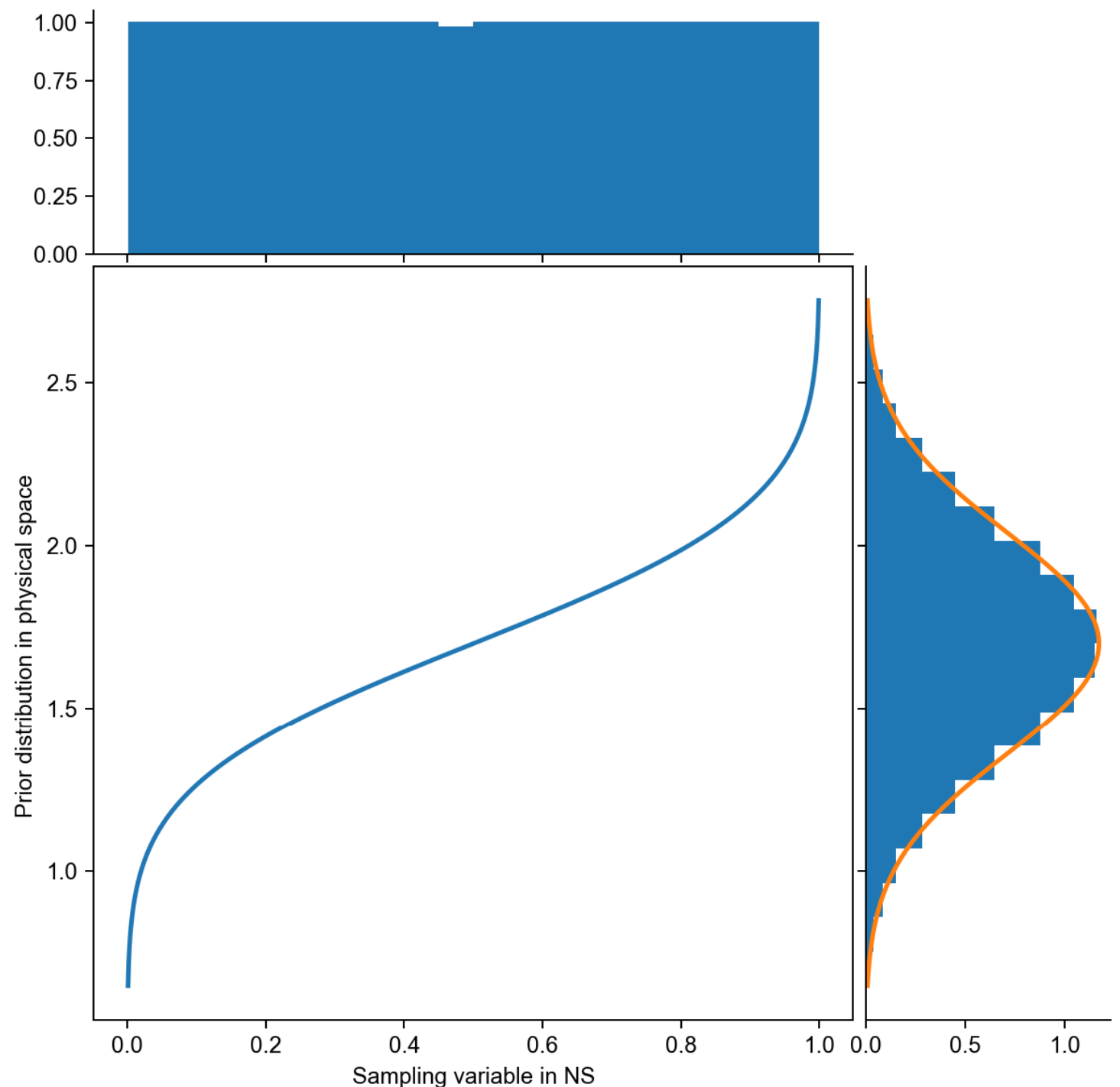
This hypercube is
(confusely) called *prior*

Model parameters must be
converted from the unit hypercube
to physical values

Uniform → rescaling

Analytical priors → inverse of the
cumulative distribution function

Strange priors → ???

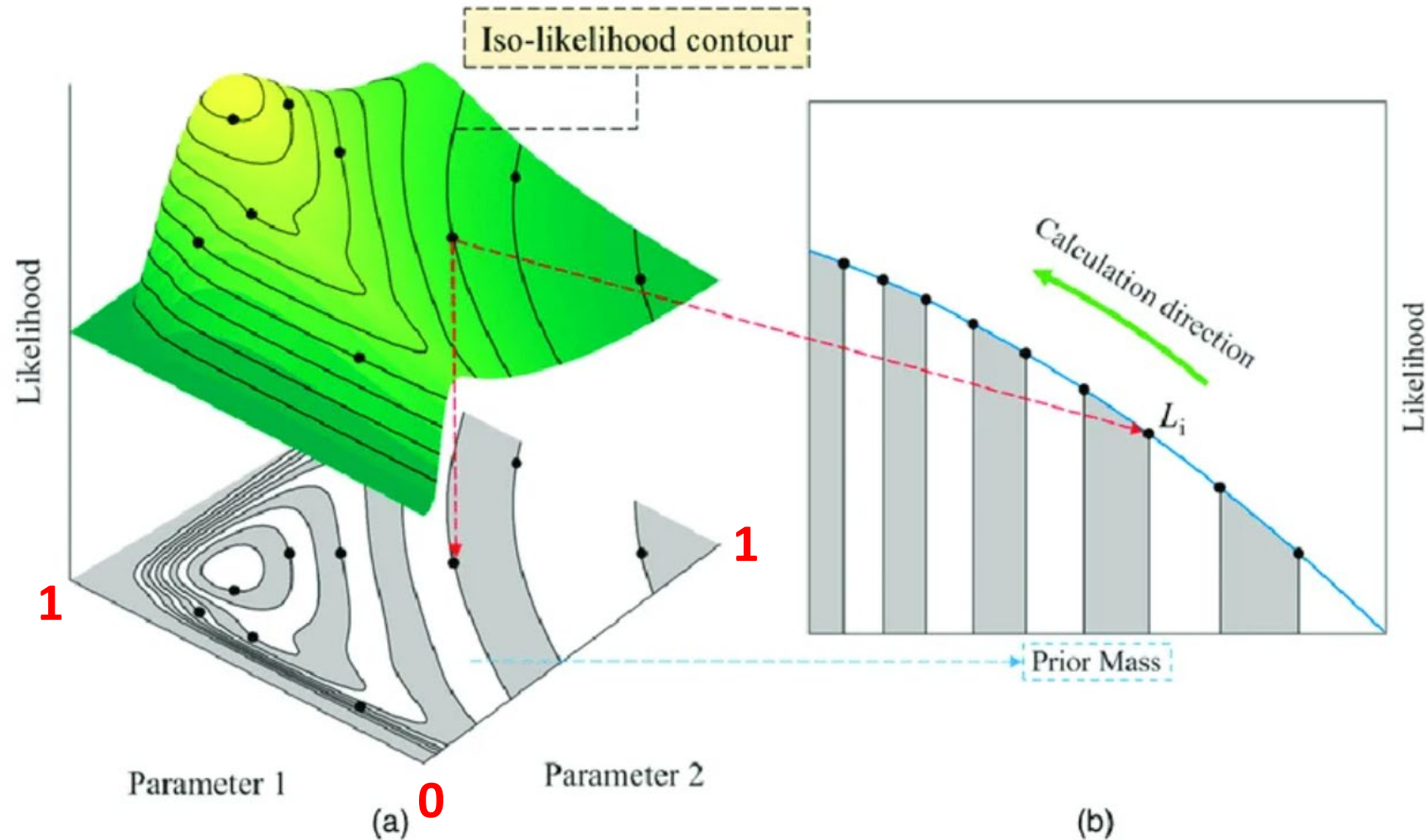


Nested sampling

Parameters X are uniformly sampled in the range $[0,1]$:
unit hypercube

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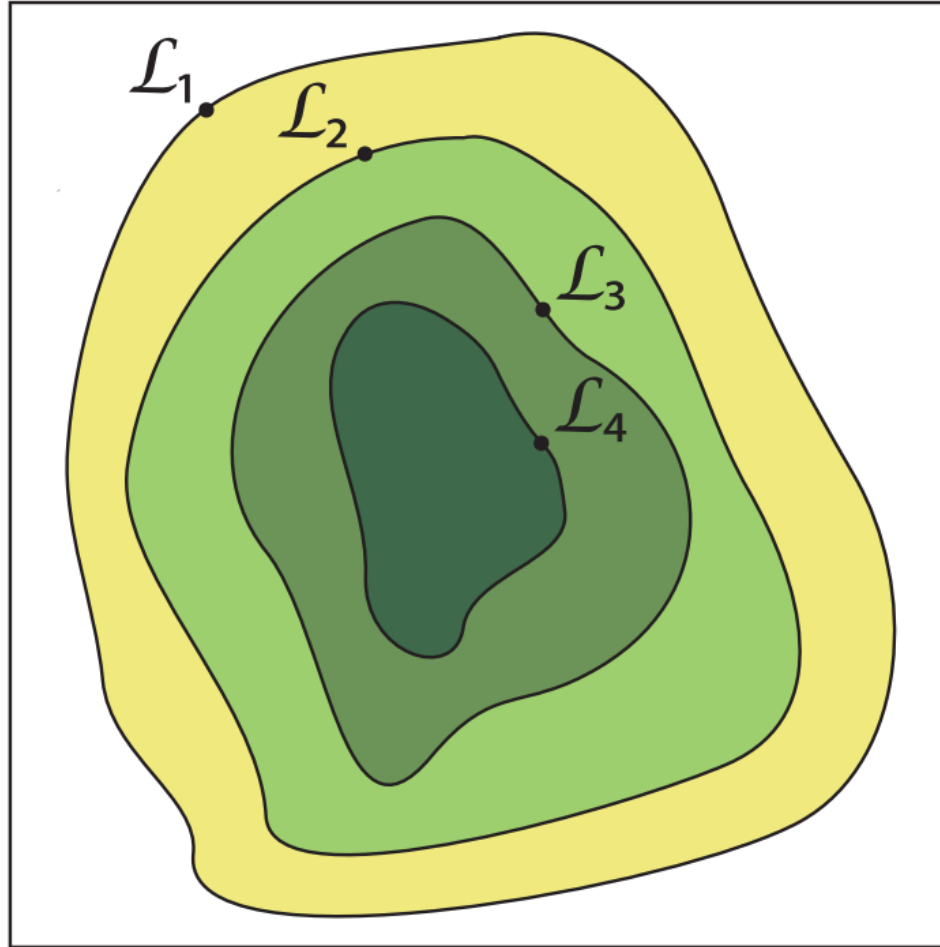
The goal is to minimise the
prior volume encompassing
the **region** with the **highest**
likelihood: **posterior mass**



This is done by randomly extracting set of parameters from the hypercube: **live points**

The live point with the lowest $\mathcal{L}(X_{\text{old}})$ is removed from the same, and a new X_{new} is generated and accepted if $\mathcal{L}(X_{\text{new}}) > \mathcal{L}(X_{\text{old}})$

Nested sampling



(a)

Prior reduction by selecting regions of the parameter space with same $\mathcal{L}_i$

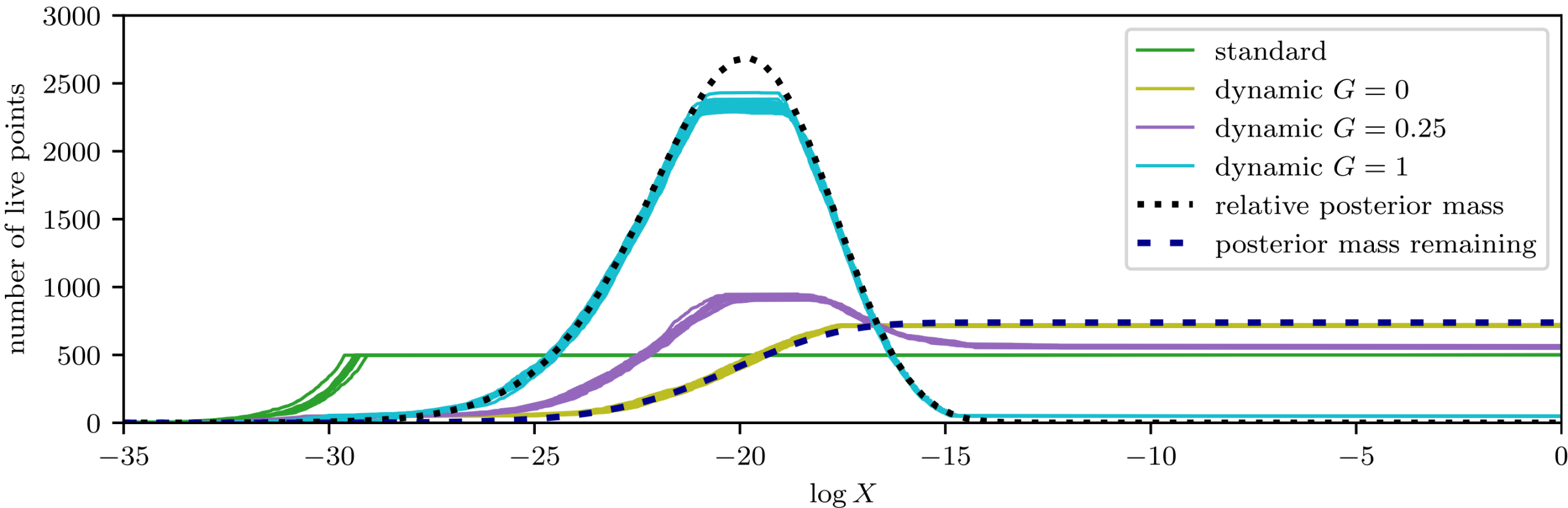
(a) the posterior of a two dimensional problem; and (b) the transformed $\mathcal{L}(X)$ function where the prior volumes X_i are associated with each likelihood $\mathcal{L}_i$

parameter space with same $\mathcal{L}_i$

; and (b) the transformed $\mathcal{L}(X)$ function

for each likelihood $\mathcal{L}_i$

Dynamic Nested Sampling

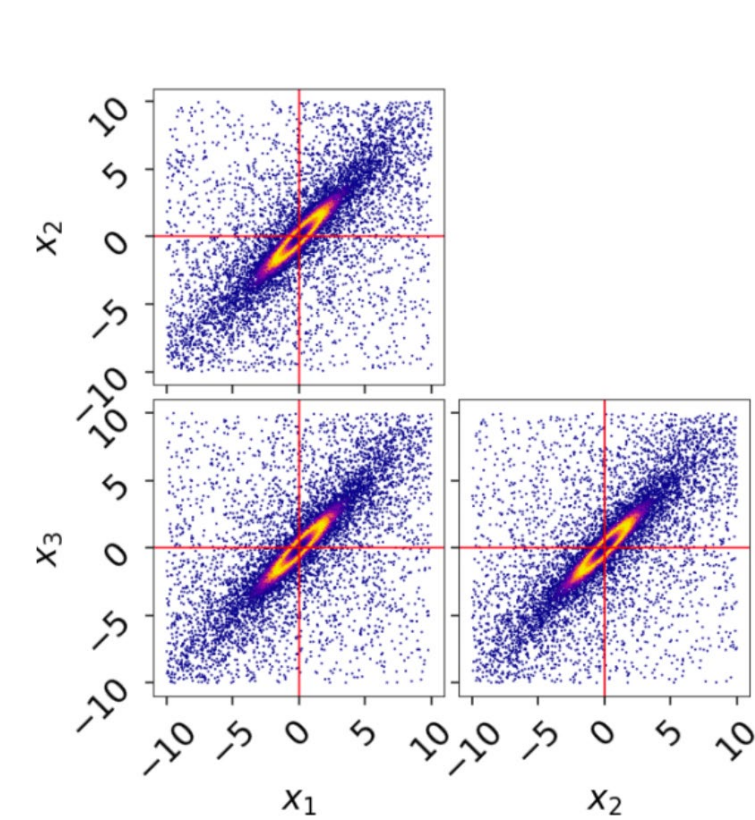


Time (iterations) goes from right to left!

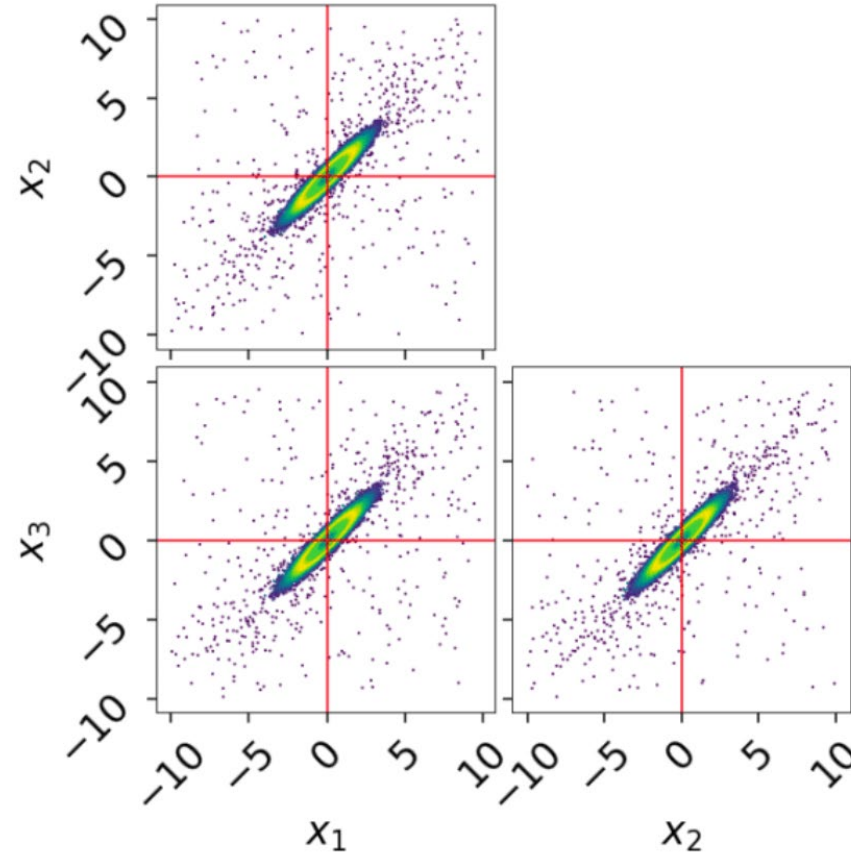
The number of live points increases as the region with the majority of the **posterior mass** is identified, and the **prior volume** decreases

Dynamic Nested Sampling: [Higson et al. 2017](#), [Speagle 2020](#)

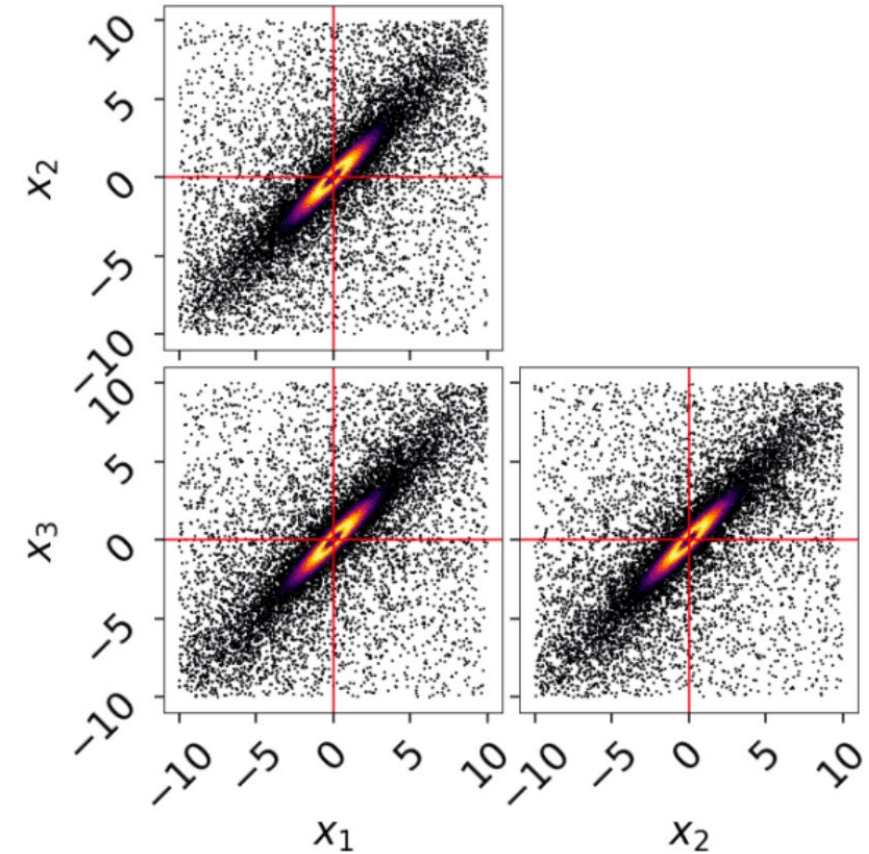
Posterior vs Prior



Static



Dynamic
(posterior)



Dynamic
(evidence)

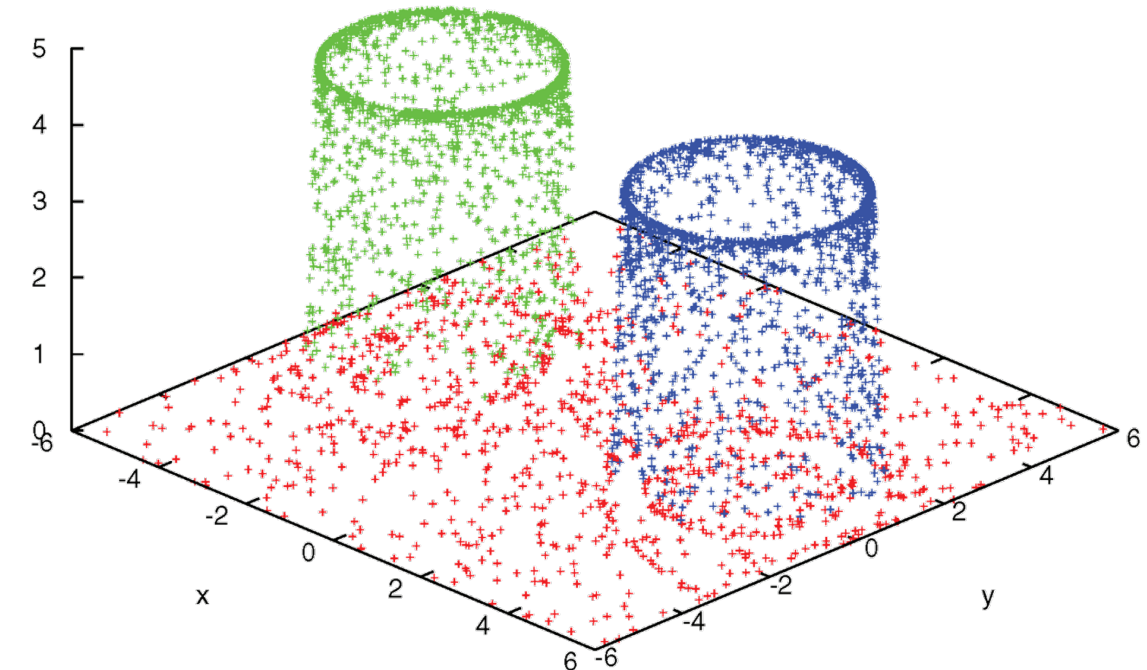
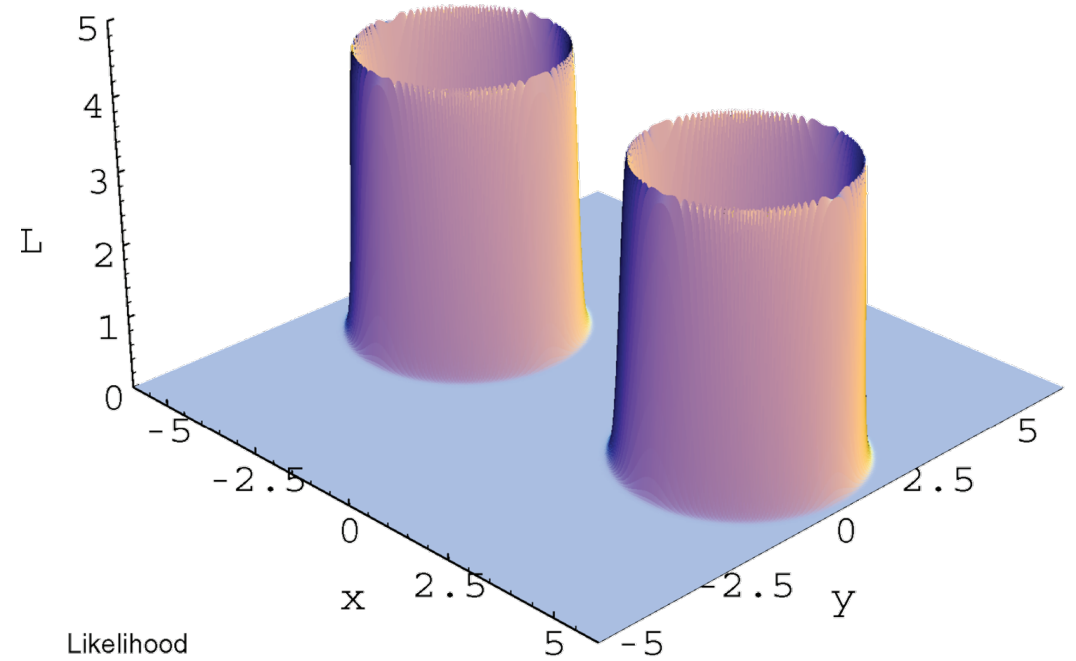
ultranest: <https://johannesbuchner.github.io/UltraNest/>

dynesty: <https://dynesty.readthedocs.io/en/latest/>

Nested Sampling

Advantages:

- No starting point required
- Exploration of the full parameter space
- **Calculation of the evidence**
- Automatic stopping criteria using evidence
- **Great for multimodal / high dimensionality problems**



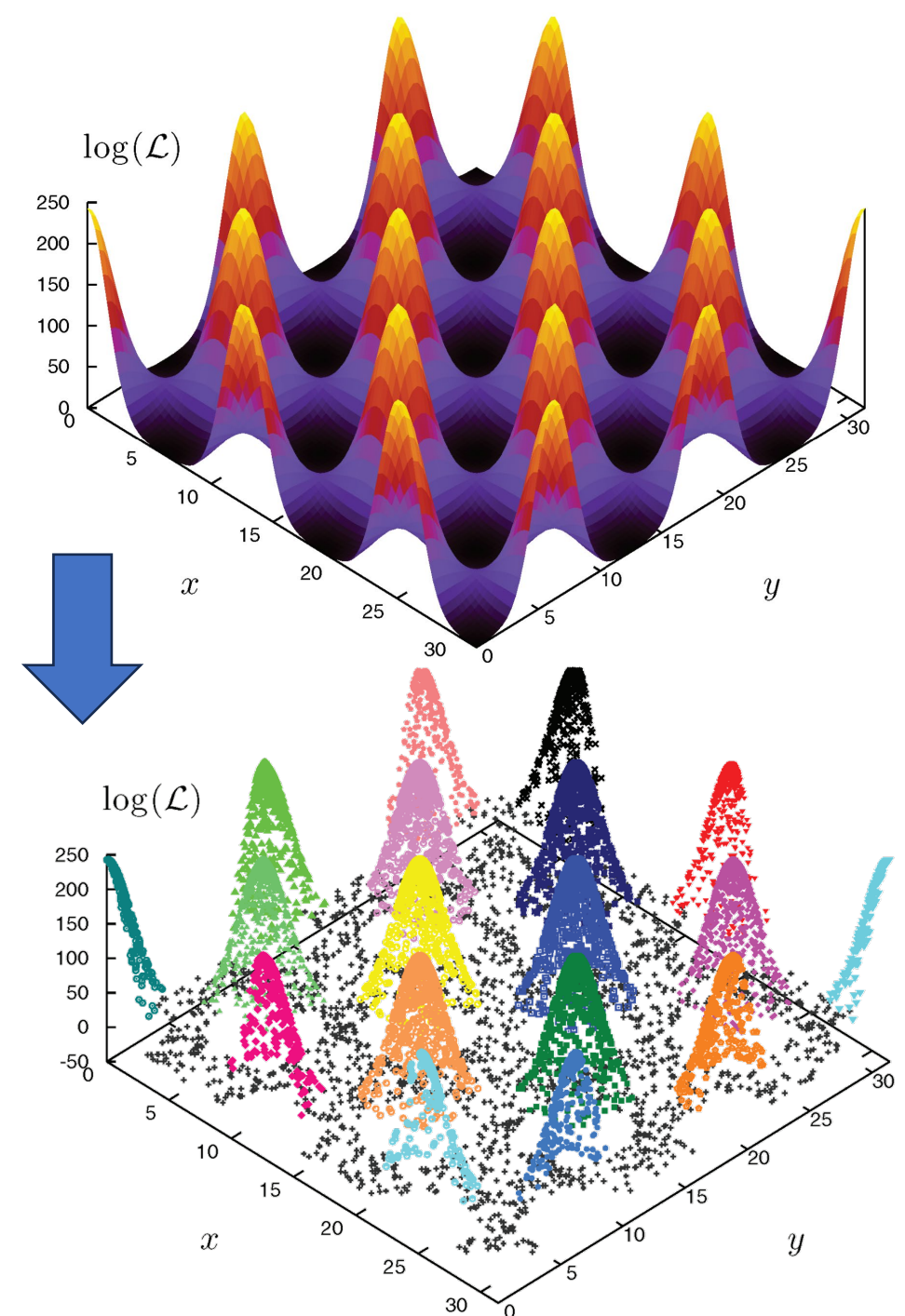
Nested Sampling

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Disadvantages:

- Difficult implementation of priors
- **Requires attention in boundaries selection**
- Computationally expensive
- **It may take much longer to collect the same number of samples around the highest likelihood region**



WHEN YOU USE MCMC FOR THE
POSTERIOR AND NS FOR THE
BAYES' FACTOR



WHEN YOU USE NS WITH
INSANELY LARGE BOUNDARIES

